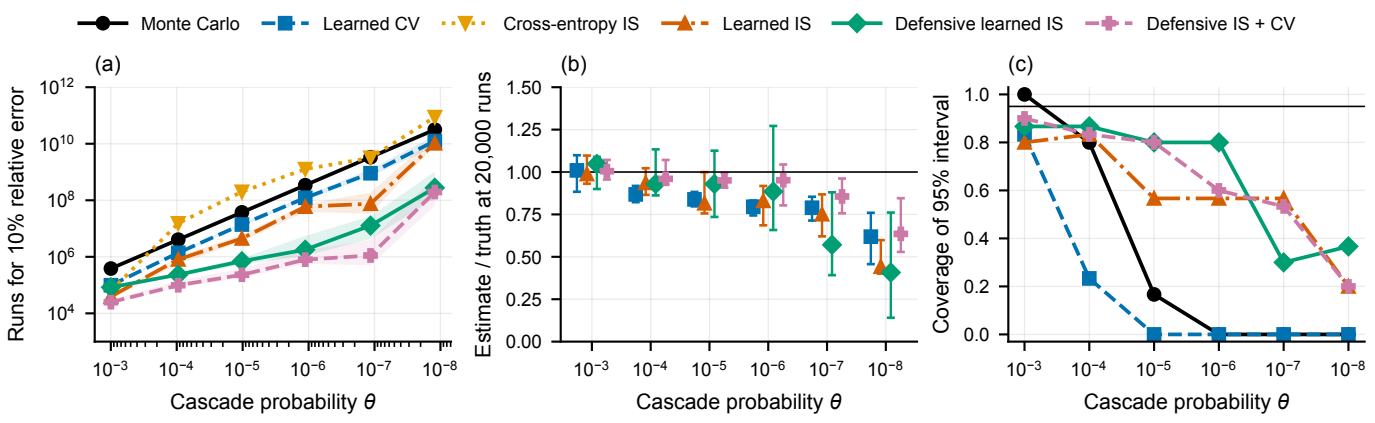
Monte Carlo
Learned CV
Cross-entropy IS
Learned IS
Defensive learned IS
Defensive IS + CV
(a)
Runs for 10% relative error
$10^{12}$
$10^{10}$
$10^{8}$
$10^{6}$
$10^{4}$
$10^{-3}$ $10^{-4}$ $10^{-5}$ $10^{-6}$ $10^{-7}$ $10^{-8}$
Cascade probability $\theta$
(b)
Estimate / truth at 20,000 runs
1.50
1.25
1.00
0.75
0.50
0.25
0.00
$10^{-3}$ $10^{-4}$ $10^{-5}$ $10^{-6}$ $10^{-7}$ $10^{-8}$
Cascade probability $\theta$
(c)
Coverage of 95% interval
1.0
0.8
0.6
0.4
0.2
0.0
$10^{-3}$ $10^{-4}$ $10^{-5}$ $10^{-6}$ $10^{-7}$ $10^{-8}$
Cascade probability $\theta$